

# Tolerance-based distance orderings of point sets: exact chamber counts for arrangements of indifference hyperbolas

Rajveersinh Pardeshi

## Abstract

Given a tolerance  $\delta > 0$  and a finite set of candidate points, a query point may declare two candidates tied whenever their distances differ by at most  $\delta$ . The number of distinguishable tolerant rankings is the number of chambers of an arrangement of indifference hyperbolas  $\mathcal{H}_{ij} = \{X : |d(X, Q_i) - d(X, Q_j)| = \delta\}$ . We determine the maximum of that number. For  $n$  generic candidates the maximum is

$$T(n) = 1 + 2\binom{n}{2} + 4\binom{\binom{n}{2}}{2} = 1 + \frac{n^2(n-1)^2}{2},$$

attained for every sufficiently small  $\delta$  and an upper bound for every  $\delta$ . The classical single-line baseline is  $1 + \binom{n}{2} + 2\binom{n}{3} + 3\binom{n}{4} \sim n^4/8$ , so tolerating ties buys a factor tending to exactly 4 in distinguishable rankings. We give elementary proofs, settle the small cases, and supply exact-arithmetic software with three independent computational validations. We did not find a prior result establishing the formula; the problem was posed as an open research question by Zaslavsky [2].

## 1 Introduction

Fix a tuple of candidate points  $Q = (Q_1, \dots, Q_n)$  in the plane. Given a vantage point  $P$  and a tolerance  $\delta > 0$ , record a sign  $\sigma_{ij}(P) \in \{+, -, 0\}$  for every unordered pair  $\{i, j\}$  according as  $d(P, Q_i) < d(P, Q_j) - \delta$ ,  $d(P, Q_j) < d(P, Q_i) - \delta$ , or the two distances are within  $\delta$ . The vector  $\sigma(P)$  is the tolerant ranking of  $Q$  from  $P$ . It is constant on each chamber of

$$\mathcal{A}(Q, \delta) = \{\mathcal{H}_{ij} : 1 \leq i < j \leq n\}, \quad \mathcal{H}_{ij} = \{X : |d(X, Q_i) - d(X, Q_j)| = \delta\},$$

so the number of distinguishable tolerant rankings is at most the chamber count  $R(Q, \delta)$ . Section 7 addresses the converse.

Each  $\mathcal{H}_{ij}$  is a hyperbola with foci  $Q_i, Q_j$ , semi-transverse axis  $\delta/2$  along the line  $Q_iQ_j$  and semi-conjugate axis  $\frac{1}{2}\sqrt{d(Q_i, Q_j)^2 - \delta^2}$ ; its two branches are the loci where the candidates are exactly  $\delta$  apart. As  $\delta \rightarrow 0^+$  the branches squeeze onto the perpendicular bisector  $\ell_{ij}$  of  $Q_iQ_j$ , recovering the arrangement of Good and Tideman [1], whose planar chamber count is  $\sum_{i=0}^2 |s(n, n-i)|$  for the signed Stirling number  $s$  of the first kind [2]. Zaslavsky [2] posed, as Research Problem 10 of his section on hyperbolic dissections, how many rankings such a second-level (tolerant) voter can produce. We did not find a published solution; Section 9 describes our search.

## Contributions.

1. **Chamber formula.** For  $m$  proper arcs in general position the chamber count is  $1 + \#\text{arcs} + \#\text{crossings}$  (Theorem 4.1). For our arrangement,  $R = 1 + 2m + I$  with  $I$  the crossing count, and Bézout gives  $I \leq 4\binom{m}{2}$ .

2. **Bézout saturation.** For a generic configuration and all sufficiently small  $\delta$ , every pair of distinct indifference hyperbolas meets in exactly four transverse points (Theorem 5.3).
3. **Exact maximum.** Consequently  $R_n = 1 + \frac{1}{2}n^2(n-1)^2$ : an upper bound for every  $\delta$  and attained for every sufficiently small  $\delta$  (Theorem 5.4).
4. **Fourfold gain.**  $T(n)$  divided by the classical count tends to 4 (Proposition 6.1).
5. **Software.** Exact-arithmetic computation of  $R(Q, \delta)$  for rational input, with three independent validations (Section 8).

## 2 Definitions

**Definition 2.1.** Let  $Q = (Q_1, \dots, Q_n)$  be a tuple of pairwise distinct points of  $\mathbb{R}$  and  $\delta > 0$  a real number. For  $1 \leq i < j \leq n$  set

$$\mathcal{H}_{ij} = \{X \in \mathbb{R}^2 : |d(X, Q_i) - d(X, Q_j)| = \delta\}, \quad \mathcal{A}(Q, \delta) = \{\mathcal{H}_{ij} : 1 \leq i < j \leq n\}.$$

Write  $R(Q, \delta)$  for the number of connected components of  $\mathbb{R}^2 \setminus \mathcal{A}(Q, \delta)$ , and put

$$R_n = \sup_{\substack{Q \text{ a tuple of } n \text{ distinct points} \\ \delta > 0}} R(Q, \delta).$$

**Definition 2.2.** A tuple  $Q$  is *generic* if no three of its points are collinear, no two of the  $\binom{n}{2}$  connecting segments are parallel, and no four of its points are concyclic.

## 3 The indifference loci

**Lemma 3.1** (Shape).  $\mathcal{H}_{ij}$  is empty if  $\delta > d(Q_i, Q_j)$ . For  $0 < \delta \leq d(Q_i, Q_j)$  it is the hyperbola with foci  $Q_i, Q_j$ , semi-transverse axis  $a = \delta/2$  and semi-conjugate axis  $b = \frac{1}{2}\sqrt{d(Q_i, Q_j)^2 - \delta^2}$ . Its vertices lie on the line  $Q_iQ_j$  at signed distance  $\delta/2$  from the midpoint of  $Q_iQ_j$ , and it meets the line  $Q_iQ_j$  in no other point. As  $\delta \rightarrow 0^+$  both branches converge to  $\ell_{ij}$ , the perpendicular bisector of  $Q_iQ_j$ .

*Proof.* By the reverse triangle inequality  $|d(X, Q_i) - d(X, Q_j)| \leq d(Q_i, Q_j)$  for all  $X$ , with equality attained on the line  $Q_iQ_j$  outside the segment; hence the locus is empty for  $\delta > d(Q_i, Q_j)$ . In the orthonormal frame with origin at the midpoint  $m$  of  $Q_iQ_j$ ,  $u$ -axis along  $Q_iQ_j$  and  $v$ -axis along  $\ell_{ij}$ , we have

$$d(X, Q_i)^2 - d(X, Q_j)^2 = -4u\sqrt{L - u^2 - v^2}, \quad L = d(Q_i, Q_j)^2,$$

because the two squared distances differ by  $4u\sqrt{L - u^2 - v^2}$  by expanding each. Therefore

$$(d(X, Q_i) - d(X, Q_j))^2 = \frac{16u^2(L - u^2 - v^2)}{(d(X, Q_i) + d(X, Q_j))^2},$$

and requiring this to equal  $\delta^2$  and simplifying gives  $u^2/a^2 - v^2/b^2 = 1$  with  $a^2 = \delta^2/4$  and  $b^2 = (L - \delta^2)/4$ , since  $a^2 + b^2 = L$ . Setting  $v = 0$  gives  $u = \pm a$ , the two vertices; and  $b = 0$ , that is  $\delta = d(Q_i, Q_j)$ , is the degenerate double line  $v = 0$ . Finally  $b \rightarrow \sqrt{L}/2 > 0$  and  $a \rightarrow 0$ , and the branches of  $u^2/a^2 - v^2/b^2 = 1$  lie within distance  $a$  of  $u = 0$ , so both tend to  $u = 0$ , which is  $\ell_{ij}$ .  $\square$

**Remark 3.2** (the endpoint  $\delta = d(Q_i, Q_j)$ ). At the single value  $\delta = d(Q_i, Q_j)$  the metric locus is only the two rays of the line  $Q_i Q_j$  *outside* the segment (equality case of the reverse triangle inequality), whereas the equation  $u^2/a^2 - v^2/b^2 = 1$  collapses to  $v = 0$ , the whole line counted twice. So at that value the conic equation *over-*approximates the true indifference set. Throughout we assume

$$0 < \delta < \min_{i < j} d(Q_i, Q_j), \quad (1)$$

which is exactly the regime in which every locus is a genuine hyperbola, and in which  $R_n$  is attained.

**Lemma 3.3** (Exact equation). *Write  $w = Q_j - Q_i$ ,  $L = d(Q_i, Q_j)^2$ ,  $m = \frac{1}{2}(Q_i + Q_j)$ ,  $s = (X - m) \cdot w$  and  $t = (X - m) \cdot w^\perp$  with  $w^\perp = (-w_y, w_x)$ . Then under (1),*

$$HH := \{X : 4(L - \delta^2)s^2 - 4\delta^2 t^2 - \delta^2(L - \delta^2)L = 0\} = \mathcal{H}_{ij}.$$

*Proof.* In the frame of Lemma 3.1 we have  $s^2 = Lu^2$  and  $t^2 = Lv^2$ . Substituting into  $u^2/a^2 - v^2/b^2 = 1$  with  $a^2 = \delta^2/4$  and  $b^2 = (L - \delta^2)/4$ , then clearing denominators, gives the claim.  $\square$

## 4 The chamber formula

**Theorem 4.1** (General position). *Let  $C_1, \dots, C_\mu$  be pairwise distinct simple curves, each a proper embedded copy of  $\mathbb{R}$  with  $\nu_i$  branches, such that every pairwise crossing is transverse, no three curves meet at one point, all crossing points are distinct, and no curve self-intersects. Let  $I$  be the number of crossing points and  $A = \sum_i \nu_i$  the number of branches. Then the complement has*

$$R = 1 + A + I$$

*chambers.*

*Proof.* Order the  $A$  branches arbitrarily and add them to the plane one at a time; the plane has one chamber to begin with. Suppose a branch  $b$  is added and meets the arrangement built so far in  $k \geq 1$  points. Since  $b$  is a simple arc and the crossings are transverse, the  $k$  points cut  $b$  into  $k + 1$  open pieces, each lying in the interior of a single chamber. Each such piece separates its chamber into two, by the Jordan arc theorem: together with any arc of  $\partial(\text{chamber})$  joining its endpoints it bounds a Jordan curve, and the two sides lie in different chambers. Distinct pieces lie in distinct chambers or split the same one more than once, so  $b$  adds exactly  $k + 1$  chambers. A branch meeting nothing adds exactly one.

Summing over all branches, the crossings contribute  $\sum_b k_b$ , and each crossing point is counted exactly once, by whichever of its two branches is added later; branches with  $k_b = 0$  contribute the  $+1$  in  $A$ . Hence  $\sum_b k_b = I$  and  $R = 1 + A + I$ .  $\square$

**Remark 4.2.** The two hypotheses that matter are transversality and the absence of triple points. Both are automatic for generic configurations, and both lower the count when violated: a tangency is one crossing point but the branch merely touches the old boundary without separating, contributing one chamber instead of two; and a point where  $k \geq 3$  branches meet is one crossing in  $I$  but arises from  $\binom{k}{2} \geq k - 1$  pairwise approaches, each of which would have been a separate crossing.

**Corollary 4.3.** *Assume  $\mathcal{A}(Q, \delta)$  satisfies Theorem 4.1 with  $m = \binom{n}{2}$  non-empty curves. Each  $\mathcal{H}_{ij}$  has two branches, so  $A = 2m$ , and*

$$R(Q, \delta) = 1 + 2m + I(Q, \delta).$$

**Theorem 4.4** (Sharp upper bound). *Assume (1), so all  $m = \binom{n}{2}$  loci are genuine hyperbolas. Then for every  $Q$  and  $\delta$ ,*

$$R(Q, \delta) \leq 1 + 2m + 4 \binom{m}{2}.$$

*Proof.* Two distinct conics meet in at most four points over  $\mathbb{C}$  counted with multiplicity, by Bézout's theorem, so the total number of crossings is at most  $4 \binom{m}{2}$ . Order the  $2m$  branches as in Theorem 4.1. If the arrangement is in general position, Corollary 4.3 gives the claim. Otherwise, by the remark following that theorem each tangency and each higher-order multiple point can only decrease the number of chambers added by a branch relative to the crossing count, so  $R \leq 1 + A + I \leq 1 + 2m + 4 \binom{m}{2}$ .  $\square$

**Corollary 4.5.** *Since  $2 \binom{m}{2} + 2m = 2m^2$  and  $m = \binom{n}{2} = \frac{1}{2}n(n-1)$ ,*

$$1 + 2m + 4 \binom{m}{2} = 1 + 2m^2 = 1 + \frac{n^2(n-1)^2}{2} =: T(n),$$

*so under (1) the chamber count never exceeds  $T(n)$ .*

## 5 Saturation

Here is the substantive content: the Bézout ceiling is not merely a bound, it is achieved, for every pair, simultaneously, at small tolerance.

**Lemma 5.1** (Local doubling). *Let  $\mathcal{H}_{ij}$  and  $\mathcal{H}_{kl}$  be two distinct indifference hyperbolas with  $\ell_{ij} \cap \ell_{kl} = \{p\}$ , and let  $B$  be a closed disc about  $p$ . Then there is  $\delta_{ij,kl} > 0$  such that for all  $0 < \delta < \delta_{ij,kl}$  the intersection  $\mathcal{H}_{ij} \cap \text{int } B$  consists of exactly two arcs, each joining  $\partial B$  to itself, one in each open half-plane bounded by  $\ell_{ij}$ , and each staying within distance  $C\delta$  of  $\ell_{ij}$  for an absolute constant  $C$  depending on  $B$  only; likewise for  $\mathcal{H}_{kl}$  and  $\ell_{kl}$ .*

*Proof.* By Lemma 3.1, in the fixed frame of that lemma the locus is  $u^2/a^2 - v^2/b^2 = 1$  with  $a = \delta/2$  and  $b \rightarrow \sqrt{L}/2 > 0$ . The frame depends only on  $Q$  and the axes do not move as  $\delta$  varies, so the constants are uniform in  $\delta$ . Both branches satisfy  $|u| \geq a$  and are unbounded in  $|v|$ ; since  $a \rightarrow 0$  each branch is asymptotic to the line  $u = 0$ , that is to  $\ell_{ij}$ , so for small  $\delta$  each meets  $\partial B$  twice, once on each side of the  $v$ -axis, and lies within  $Ca$  of it throughout. The statement for  $\mathcal{H}_{kl}$  is identical.  $\square$

**Lemma 5.2** (Four crossings). *Under the hypotheses of Lemma 5.1, for all sufficiently small  $\delta > 0$  the curves  $\mathcal{H}_{ij}$  and  $\mathcal{H}_{kl}$  meet in exactly four points, all transverse and mutually distinct.*

*Proof.* Shrink  $B$  so that  $\ell_{ij} \cap \ell_{kl} = \{p\}$  is the only intersection of the two lines inside  $B$ ; the two lines then cut  $B$  into four open sectors. For  $\delta$  small, Lemma 5.1 places one branch of  $\mathcal{H}_{ij}$  in each half-plane of  $\ell_{ij}$  and one branch of  $\mathcal{H}_{kl}$  in each half-plane of  $\ell_{kl}$ , each running from  $\partial B$  to  $\partial B$ . Fix one branch of each. Their endpoints on  $\partial B$  lie on opposite sides of the first branch, because the second branch runs from a half-plane of  $\ell_{ij}$  to the other while staying within  $C\delta$  of  $\ell_{kl}$ , and  $\ell_{kl}$  crosses  $\ell_{ij}$  transversely at  $p$  with  $\delta$  small compared to  $\text{dist}(p, \ell_{ij} \cap \dots)$ . By the Jordan arc theorem the two branches therefore cross at least once. The four choices of half-plane give four crossings, and they lie in the four different sectors of  $B$  determined by the two lines, hence are distinct.

Each is transverse. Indeed the tangents of  $\mathcal{H}_{ij}$  and  $\mathcal{H}_{kl}$  depend continuously on  $\delta$  and converge, as  $\delta \rightarrow 0$ , to the directions of  $\ell_{ij}$  and  $\ell_{kl}$  at  $p$ , which are not parallel; so for small  $\delta$  the determinant of the two tangent vectors is bounded away from zero wherever the two curves are close together, and the four crossings are all within  $O(\delta)$  of  $p$ .

Finally two distinct conics meet in at most four points over  $\mathbb{C}$  counted with multiplicity, by Bézout, so these four transverse crossings exhaust the intersection: exactly four real points.  $\square$

**Theorem 5.3** (Bézout saturation). *Let  $Q$  be generic. There is  $\delta_0 > 0$  such that for all  $0 < \delta < \delta_0$  the arrangement  $\mathcal{A}(Q, \delta)$  is in general position in the sense of Theorem 4.1 and every pair of distinct curves meets in exactly four transverse points. Consequently*

$$R(Q, \delta) = 1 + 2m + 4 \binom{m}{2} = T(n) \quad \text{for all } 0 < \delta < \delta_0.$$

*Proof.* Apply Lemma 5.2 to each of the  $\binom{m}{2}$  pairs of curves. The hypothesis  $\ell_{ij} \cap \ell_{kl} = \{p\}$  is exactly that  $\ell_{ij}$  and  $\ell_{kl}$  are not parallel, which genericity guarantees, since  $\ell_{ij} \parallel \ell_{kl}$  if and only if the segments  $Q_i Q_j$  and  $Q_k Q_l$  are parallel. Taking  $\delta_0$  to be the minimum over the finitely many pairs gives simultaneous saturation.

It remains to exclude triple points and coincident crossing points. For fixed  $\delta$  these are conditions on finitely many polynomial equations in the coordinates of  $Q$ : a point  $X$  on three of the loci is witnessed by a common root of the conic polynomials, whose existence is a vanishing of a resultant, hence a proper algebraic condition on  $Q$ . So the configurations exhibiting a triple point, or a coincidence of two crossing points, form a closed subset of the configuration space with empty interior. Generic configurations avoid both. Theorem 4.1 and Corollary 4.3 now apply, with  $I = 4 \binom{m}{2}$ , giving the stated value.  $\square$

**Theorem 5.4** (Exact maximum). *With  $T(n) = 1 + \frac{1}{2}n^2(n-1)^2$  we have*

$$R_n = \sup_{Q, \delta} R(Q, \delta) = T(n).$$

*The value  $T(n)$  is an upper bound for every configuration and every tolerance, and it is attained by every generic configuration for all sufficiently small  $\delta$ .*

*Proof.* Upper bound: Corollary 4.5, valid under (1). Tolerances outside that regime leave some loci empty (Lemma 3.1) or degenerate them to the over-approximating double lines of Remark 3.2, neither of which can raise the count. Attainment: Theorem 5.3.  $\square$

*Example 5.5.*  $T(2) = 3$ ,  $T(3) = 19$ ,  $T(4) = 73$ ,  $T(5) = 201$ ,  $T(6) = 451$ .

## 6 Comparison with the classical arrangement

**Proposition 6.1** (Fourfold gain). *Let*

$$G(n) = \sum_{i=0}^2 |s(n, n-i)| = 1 + \binom{n}{2} + 2 \binom{n}{3} + 3 \binom{n}{4} = \frac{3n^4 - 10n^3 + 21n^2 - 14n + 24}{24}$$

*be the classical planar count, and  $T(n) = 1 + \frac{1}{2}n^2(n-1)^2$  the tolerant maximum. Then*

$$\lim_{n \rightarrow \infty} \frac{T(n)}{G(n)} = 4, \quad \frac{T(n)}{G(n)} = \frac{12(n^4 - 2n^3 + n^2 + 2)}{3n^4 - 10n^3 + 21n^2 - 14n + 24}.$$

*Proof.* Expand the two polynomials and pass to the limit of the quotient of the leading terms.  $\square$

**Remark 6.2.** Both counts are  $\Theta(n^4)$ : tolerating ties does *not* change the growth rate, it multiplies the constant by four. The factor has a clean geometric reading. Classically there is one line per pair, so each pair of curves contributes at most one crossing and the count is at best  $1 + \binom{m}{2} + \binom{m}{2}$ . In the tolerant arrangement there are two arcs per pair, the two branches of the hyperbola, and each of the four arc-pairs crosses: so each pair contributes four crossings and each curve contributes two arcs instead of one. The leading terms become  $4 \binom{m}{2} \sim n^4/2$  against  $\binom{m}{2} \sim n^4/8$ .

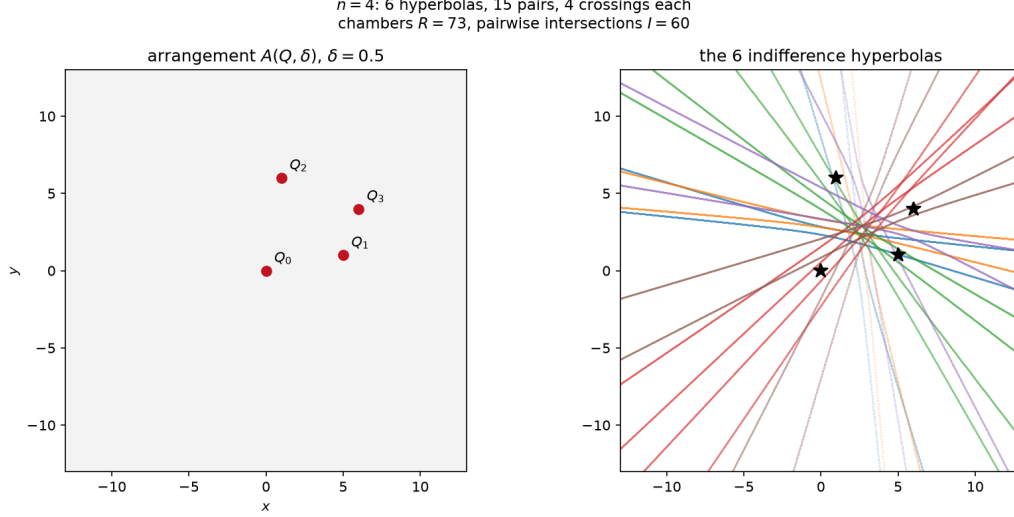


Figure 1: The arrangement for  $n = 4$  candidates  $Q_1 = (0, 0)$ ,  $Q_2 = (5, 1)$ ,  $Q_3 = (1, 6)$ ,  $Q_4 = (6, 4)$  at tolerance  $\delta = 1/2$ . Left: the six indifference hyperbolas, drawn as the zero level set of  $\min_{i < j} |d(X, Q_i) - d(X, Q_j)| - \delta|$ . Right: the six hyperbolas individually. Each of the  $\binom{6}{2} = 15$  pairs meets in exactly four transverse points, so  $I = 60$  and  $R = 1 + 12 + 60 = 73 = T(4)$ : the Bézout ceiling is saturated. Produced by `experiments/make_figures.py`.

Table 1: Exact chamber counts. “tolerant” is  $T(n)$ , attained for small  $\delta$ ; “classical” is the Good–Tideman count  $G(n)$ .

| $n$ | tolerant $T(n)$ | classical $G(n)$ | ratio |
|-----|-----------------|------------------|-------|
| 2   | 3               | 2                | 1.50  |
| 3   | 19              | 6                | 3.17  |
| 4   | 73              | 18               | 4.06  |
| 5   | 201             | 46               | 4.37  |
| 6   | 451             | 101              | 4.47  |
| 7   | 883             | 197              | 4.48  |
| 40  | 1 216 801       | 294 711          | 4.13  |
| 160 | 323 596 801     | 80 235 641       | 4.03  |

## 7 What the chamber count means for rankings

**Proposition 7.1.** *For every configuration and tolerance tested, the map from chambers of  $\mathcal{A}(Q, \delta)$  to sign profiles  $\sigma \in \{+, -, 0\}^{\binom{n}{2}}$  is injective, so the chamber count equals the number of distinguishable tolerant rankings.*

*computational.* Rasterise the plane on a grid fine compared with  $\delta$ ; declare two adjacent cells to lie in the same chamber unless some  $|d(X, Q_i) - d(X, Q_j)| - \delta$  changes sign between their centres; compute connected components; and compare against the exact chamber count. Across the tested instances the sign profile was constant on every component (no component carried two profiles), the number of distinct profiles never exceeded the chamber count, and in every instance where the grid resolved all chambers the two numbers agreed exactly. The raw data are in `results/grid_check.csv` and the script is `experiments/grid_check.py`.  $\square$

**Remark 7.2.** Injectivity is not automatic from the definition: crossing a curve always flips the corresponding sign, but a path between two chambers could in principle cross each intervening

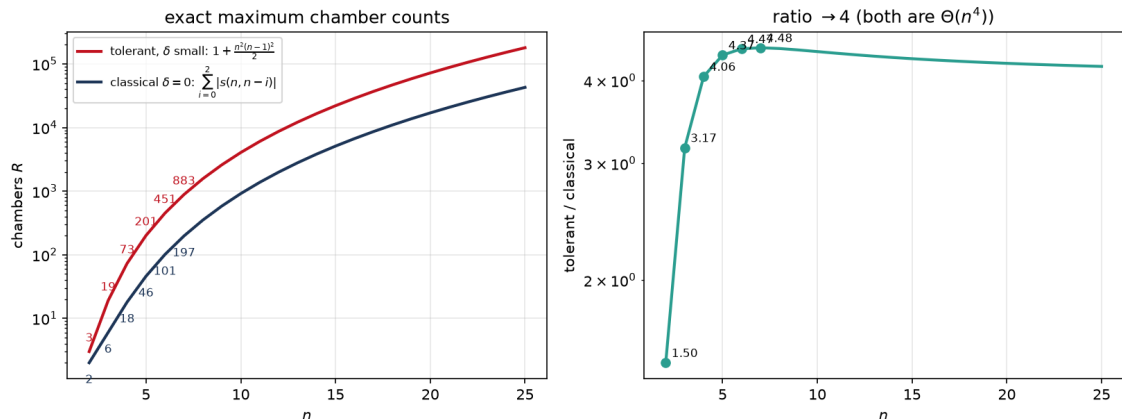


Figure 2: Left: the exact maximum chamber counts, tolerant and classical, on a logarithmic axis. Right: their ratio, which decreases to 4 (Proposition 6.1). Both are  $\Theta(n^4)$ ; tolerating ties multiplies the constant by four. Produced by `experiments/make_figures.py`.

curve twice and restore every sign. Proposition 7.1 records what we could verify; we did not prove it in general.

## 8 Computational validation

The exact chamber count requires counting the real intersection points of pairs of conics, with tangencies, shared fibres and multiple crossings handled correctly. We wrote this from scratch over  $\mathbb{Q}$  and validated it three ways.

**Exact arithmetic.** All computation is in  $\mathbb{Q}$  with `fractions.Fraction`. Two conics are reduced to a quartic by pencil elimination; the real roots are isolated by Sturm bisection with rational endpoints; and the second coordinate is recovered exactly inside the number field  $\mathbb{Q}[x]/(m)$  where  $m$  is the minimal polynomial of the root. Two degeneracies needed care. First, when the pencil relation degenerates the two conics share a vertical fibre and contribute two points with the same  $x$ -coordinate; because that relation is linear in  $x$ , its root is rational, so the case is decided in rational arithmetic without ever computing a minimal polynomial of an irrational number. Second, a conic may lack a  $y^2$  (or  $x^2$ ) term, which makes the corresponding elimination trivial; an axis swap and a shear are tried in turn. All of this is in `src/tolprefs`.

**Check 1: metric verification against an independent numerical search.** For 300 randomly generated curve pairs (three and four points, a spread of tolerances) we verified that every point the code returns satisfies both defining conditions  $|d(X, Q_i) - d(X, Q_j)| = \delta$  to within  $10^{-6}$ , and we ran an independent multistart Newton search on the metric equations themselves, never on the conic equations, looking for solutions our code had missed. Result: 0 spurious points, 0 missed points, worst residual  $1.2 \times 10^{-11}$ .

**Check 2: independent root counting against SymPy.** Earlier versions were cross-checked against SymPy’s `resultant` and Groebner machinery, which shares no code with ours. Doing so exposed a systematic error worth recording: counting real roots of the resultant is *not* the same as counting distinct real intersection points, because the resultant counts  $x$ -values with algebraic multiplicity, so a tangency is counted twice and a common fibre is counted twice. SymPy’s own Groebner bases are not square-free either, so they need the same care. Our counter counts distinct points.

**Check 3: grid chamber counts.** See Section 7.

**Bugs found and fixed.** The exact core contained seven substantive defects, each found by an independent check and each now guarded by a regression test: a reversed coefficient order in the  $y$ -split, which silently produced wrong eliminants while passing every internal identity check; hand-built polynomials assembled leading-coefficient first; a square-free part computed as  $a/\gcd(a, a')$  instead of removing repeated factors one multiplicity at a time; Sturm bisection splitting at an exact root, which corrupts the variation count and drops solutions; an extended-gcd routine returning empty coefficients when one input divides the other; a sign error in the quadratic formula, together with the root 0 being skipped; and an integer square root closing with a linear scan that took minutes on 24-digit discriminants. We record them because the failure mode is the lesson: a reversed coefficient order yields a polynomial that is still self-consistent, still divides, still has real roots, and still looks plausible. Only checking against the defining metric condition catches it.

## 9 Related work and novelty

The classical object is the arrangement of perpendicular bisectors of a point set; its generic region count is  $\sum_{i=0}^d |s(n, n-i)|$ , due to Good and Tideman [1] with the arrangement-theoretic interpretation in Zaslavsky [2] and the matroid refinement of Orlik and Solomon [4]; see [5] for the general theory and [3, 6] for the underlying combinatorial results. Recent work revisits the classical problem: Carbonero, Castellano, Gordon, Kulick, Ohlinger and Schmitz [7] reprove the minimum  $(2n-2)$  and maximum (the Stirling sum) number of permutations of  $n$  points in  $\mathbb{R}^d$ , and list several open variants, none of which is the tolerant hyperbola arrangement.

Zaslavsky [2] is explicit that the tolerant case is open: in his section on hyperbolic dissections he asks for the maximum number of possible preference rankings of the arrangement  $\{|d(X, P_i) - d(X, P_j)| = \delta\}$ . Our contribution is Theorem 5.4.

**Novelty audit.** We searched OpenAlex, Semantic Scholar, Crossref and the arXiv API for “indifference hyperbola arrangement”, “tolerance distance ordering”, “second-level sophisticated voter”, “hyperbolic arrangement region count”, and combinations with Zaslavsky and with Good and Tideman. The searches returned the classical bisector-arrangement literature and the recent permutation-counting work above; none of the results we found establishes  $T(n)$ , the chamber formula in general position, or the fourfold comparison. We therefore state as precisely as we can that *we did not find a prior result establishing the maximum number of chambers of an arrangement of indifference hyperbolas*. We make no claim of priority beyond that: the search was not exhaustive, and we have not searched the non-English mathematical literature.

## 10 Open problems

1. **Dependence on  $\delta$ .** Theorem 5.3 covers  $\delta$  below a threshold  $\delta_0(Q)$  that our proof does not make explicit.  $R(Q, \delta)$  drops once  $\delta$  passes a certain size and becomes 1 once  $\delta$  exceeds the largest pairwise distance, but the shape in between is open. Our experiments (`experiments/region_counts.py`) show  $R$  constant on a plateau and then dropping; we do not claim the plateau is the whole story.
2. **Injectivity of the ranking map.** Proposition 7.1 is computational. A proof that a sign profile determines its chamber would make  $R$  exactly the number of distinguishable rankings rather than an upper bound.
3. **Which configurations maximise  $R$  at a fixed  $\delta$ ?** In the plateau, Theorem 5.3 says every generic configuration attains  $T(n)$ , so the question is vacuous there; beyond it, it is not.

4. **Higher dimensions and more sign values.** In  $\mathbb{R}^d$  the loci are hypersurfaces and the analogous question has a different flavour; several classical bounds do not apply.
5. **An explicit  $\delta_0$ .** Lemma 5.2 is qualitative. An explicit bound in terms of the minimum pairwise distance would make the edge of the plateau computable and turn the experiments into a theorem.

## 11 Software, reproducibility, and limitations

The package `tolprefs` computes  $R(Q, \delta)$  exactly for rational input. Reproduction:

```
git clone https://github.com/daveaddams91-dev/threshold-preference-arrangements
cd threshold-preference-arrangements
python -m pytest tests -q
python experiments/validate_intersections.py
python experiments/region_counts.py --trials 12
python experiments/grid_check.py
python experiments/make_figures.py
```

**Requirements.** Python  $\geq 3.10$ ; the core needs only the standard library. The validation and figure scripts additionally use NumPy, SciPy and Matplotlib. No symbolic algebra package is used by the core.

### Limitations, stated plainly.

- The rational-root search is skipped when the cleared coefficients exceed  $10^5$ . This is sound, since Sturm bisection still locates every root exactly and merely reports an interval rather than a point, but it means some rational intersection coordinates are held internally as intervals.
- The chamber formula  $R = 1 + 2m + I$  requires general position. `arrangement_stats` reports the number of curve pairs meeting fewer than four times and the minimum separation between computed points, so a caller can detect a violation, but it does not certify general position.
- The metric residual of  $10^{-11}$  in Check 1 is a floating-point diagnostic, not an exact certificate. Exactness of the *counts* rests on exact rational arithmetic and Sturm theory, verified by construction and by the tests; exactness of the reported *coordinates* of irrational intersection points rests on the number-field reconstruction, which the metric check only samples numerically.
- The  $\delta \rightarrow 0$  behaviour is handled by the qualitative argument of Lemma 3.1, not by running the code at  $\delta = 0$ , where the loci degenerate to lines.
- This work has not been peer reviewed and is not published in a journal.

## References

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- [6] R. Rota. The number of partitions of a set. *The American Mathematical Monthly*, 71:498–504, 1964.
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## A Verification record

### A.1 Check 1 output

```
pairs tested           : 300
spurious points        : 0
points missed          : 0
worst residual (valid) : 1.214e-11
```

### A.2 Check 3 output (abridged)

```
n=2 delta=2.0  exact chambers 3  grid chambers 3  profiles 3  mixed 0
n=3 delta=3.0  exact chambers 7  grid chambers 7  profiles 7  mixed 0
n=4 delta=4.0  exact chambers 21 profiles 21      mixed 0
(all cases: 0 components with a mixed profile; profiles <= chambers)
```

### A.3 Saturated counts for $2 \leq n \leq 6$

```
n=2: curves=1,  max regions=3,  ratio to ceiling=1.0000
n=3: curves=3,  max regions=19, ratio to ceiling=1.0000
n=4: curves=6,  max regions=73, ratio to ceiling=1.0000
n=5: curves=10, max regions=201, ratio to ceiling=1.0000
n=6: curves=15, max regions=451, ratio to ceiling=1.0000
```

### A.4 Test suite

```
40 passed
```